ML4DD - Lipophilicity

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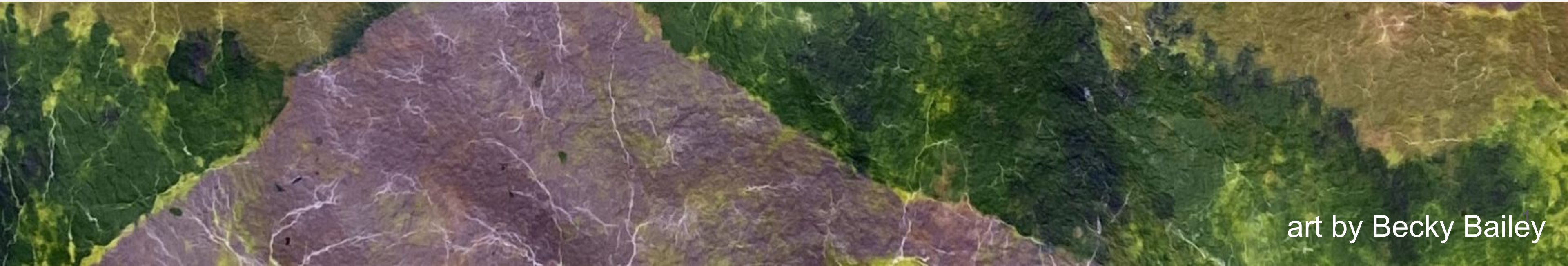


art by Becky Bailey

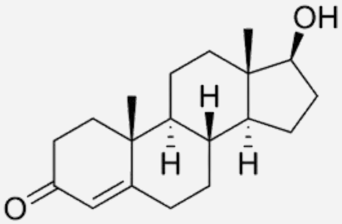
What is lipophilicity?

The tendency of a chemical to dissolve into fat (lipid) rather than water (aqueous) environments.

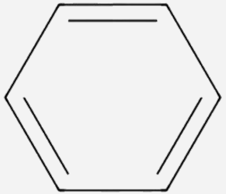
In other words molecules with high LogD are “fat-lovers”.

Lipophilicity is one of the key properties of a potential drug that determines the solubility.

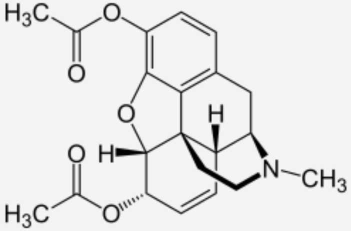
Some examples of highly lipophilic molecules:



Testosterone



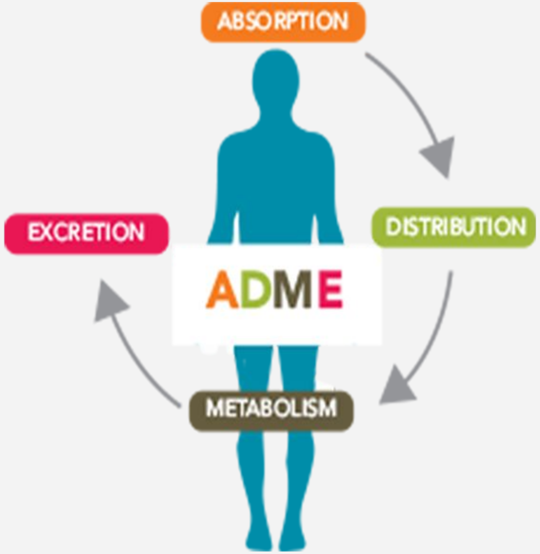
Benzene



Heroin

What is lipophilicity?

ADME properties:



-> Vital properties for any compound intended to be used as a drug, especially small molecules

First property: Absorption, the ability of a compound to travel from the site of administration to that of action

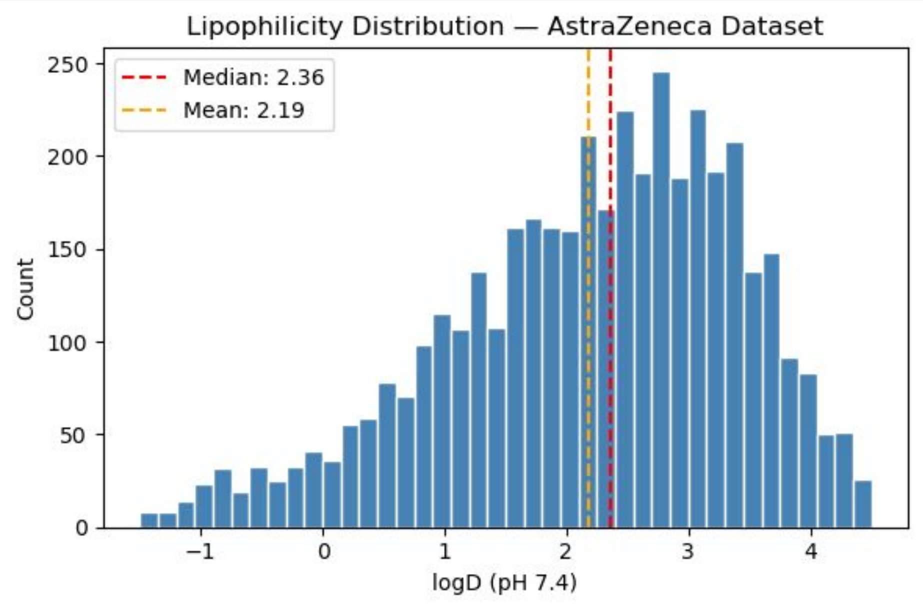
Made up by factors such as bioavailability, lipophilicity, intestinal resistance

-> Lipophilicity controls how well a compound permeates cell walls and also how well it travels in the bloodstream

Low lipophilicity == Good cell permeability, bad absorption into bloodstream, and vice versa

The Dataset

- AstraZeneca Lipophilicity dataset
- 4,200 drug-like molecules with experimentally measured logD at pH 7.4 (AstraZeneca, 2016)
- Distributed as part of the MoleculeNet benchmark suite (Wu et al., 2018)
- Regression task: predict logD from molecular structure alone
- Distribution: bell-shaped, slightly left-skewed, mean=2.19, median=2.36, range −1.5 to 4.5

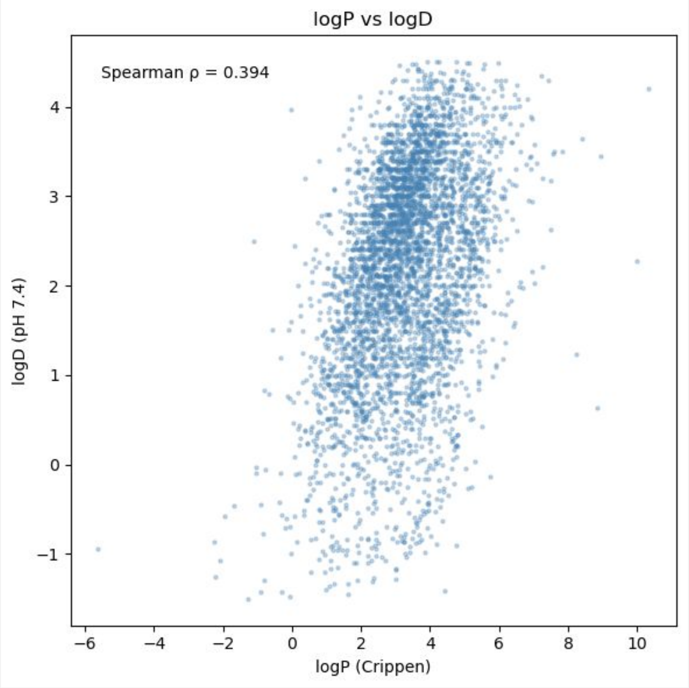


What Are We Predicting? logP vs logD

- **logP:** octanol/water partition coefficient, assumes molecule is fully neutral
- **logD at pH 7.4**: accounts for ionization at physiological pH (Avdeef, 2003)

At pH 7.4, many drugs are partially ionized → logD reflects true lipophilicity in biological conditions, logP does not

- Spearman ρ = 0.394 between logP and logD — moderate but far from perfect
- Wide vertical spread → ionization at pH 7.4 adds variability logP cannot capture
- logP alone is not sufficient → additional descriptors needed



Data Loading & Cleaning

Drug_ID	Drug	Y
CHEMBL596271	Cn1c(CN2CCN(c3ccc(Cl)cc3)CC2)nc2ccccc21	3.54
CHEMBL1951080	COc1cc(OC)c(S(=O)(=O)N2c3ccccc3CCC2C)cc1NC(=O)...	-1.18
CHEMBL1771	COC(=O)[C@H](c1ccccc1Cl)N1CCc2sccc2C1	3.69
CHEMBL317462	OC1(C#Cc2ccc(-c3ccccc3)cc2)CN2CCC1CC2	3.14
CHEMBL1940306	CS(=O)(=O)c1ccc(Oc2ccc(C#C[C@]3(O)CN4CCC3CC4)c...	1.51

```
display(df.head())
```

Dataset accessed via PyTDC (Huang et al., 2021) with three columns: `Drug_ID`, `Drug` (SMILES), `Y` (logD)

Three verification steps

Step	Method	Purpose
SMILES validity	`Chem.MolFromSmiles()`	Remove unparseable entries
Canonicalization	`Chem.MolToSmiles(canonical=True)`	Unique representation per molecule
Deduplication	Match on canonical SMILES	Remove identical molecules

Result: 0 molecules removed at every step, confirms AstraZeneca dataset is well-curated (AstraZeneca, 2016)

Molecular Representations

- Fingerprints: binary, scale-invariant → no scaling needed
- Physchem: continuous → `StandardScaler` fitted on training set only to avoid data leakage

Encoding	Bits/Features	Type	Notes
MACCS Keys	166	Binary	Predefined substructure keys (Session 2)
Morgan / ECFP4	1024	Binary	Circular, radius=2 (Session 2)
RDKit Fingerprint	2048	Binary	Path-based, maxPath=7 (Session 2)
Physchem (extended)	~100	Continuous	For feature selection (Session 6)

Splitting Strategies & Similarity Analysis

➔ **Random split** — molecules assigned randomly → structurally similar molecules in both train and test → *optimistic* baseline

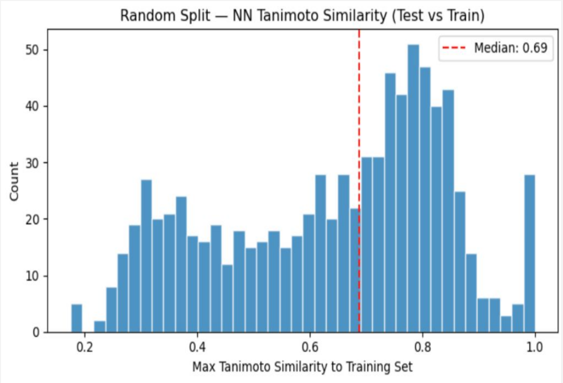
➔ **Scaffold split** — Murcko scaffold groups assigned to one partition → test molecules from unseen chemical series → *realistic* evaluation

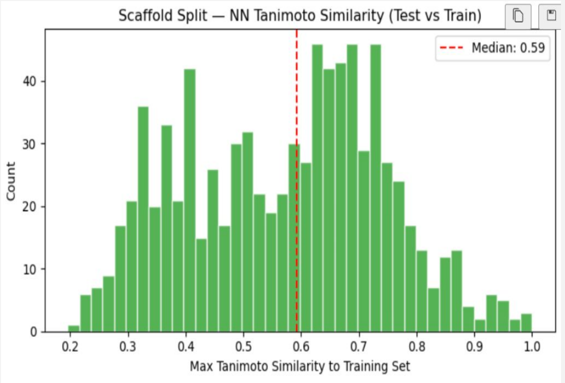
```
Random split   – Train: 2940, Val: 420, Test: 840
Scaffold split – Train: 2940, Val: 420, Test: 840
```

How different are the splits really? Nearest-neighbour Tanimoto similarity (Morgan fingerprints, Rogers & Hahn 2010):

- ~0.1 drop in median — scaffold split is harder but difference is moderate

Scaffold split = **primary evaluation**, consistent with MoleculeNet benchmark (Wu et al., 2018)

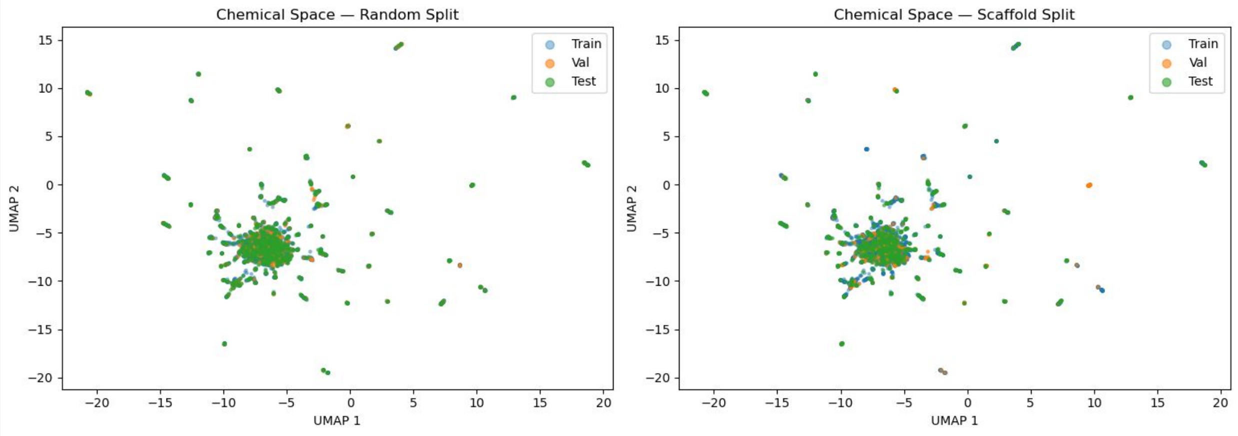




Chemical Space Visualization (UMAP)

UMAP (McInnes et al., 2018) chosen over PCA — Morgan fingerprints are high-dimensional binary vectors with non-linear structure that linear projections cannot adequately capture

- **Random split**: test set (green) dominates the entire space — train and val molecules barely visible — train and test are structurally indistinguishable
- **Scaffold split**: train (blue) and val (orange) become clearly visible inside the core alongside test molecules — more balanced structural coverage across partitions

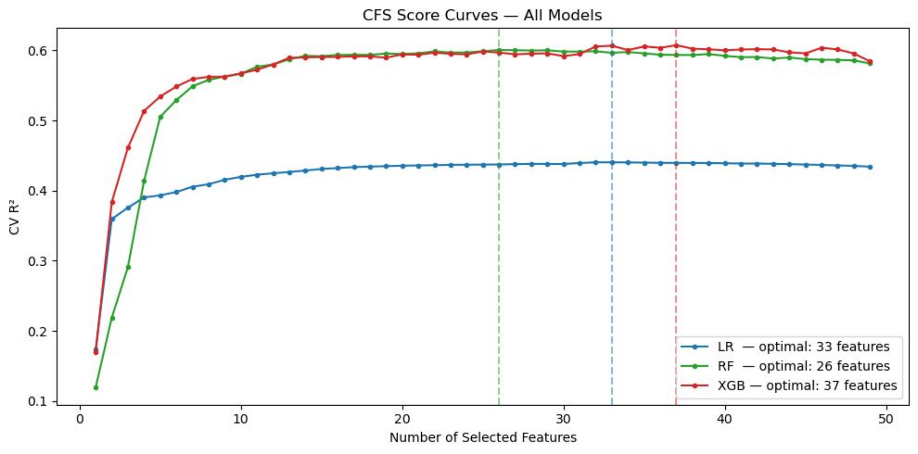


Feature Selection (Physchem)

Method	N features	R^2
All features	101	0.470
Filter	33	0.448
FS	50	0.453
CFS	33	0.452

- ~101 descriptors intentionally include correlated redundant groups — logP variants, size descriptors, connectivity indices — making feature selection essential

- Three methods on Linear Regression (baseline):
    - **Filter**
    - **Forward Selection**
    - **CFS**



- **CFS selected** — most principled, resolves substitution effect in PFI

- Applied **separately per model** since optimal features may differ depending on each model's ability to capture non-linear relationships

Full Model Comparison

3 models × 4 encodings × 2 splits = 24 combinations

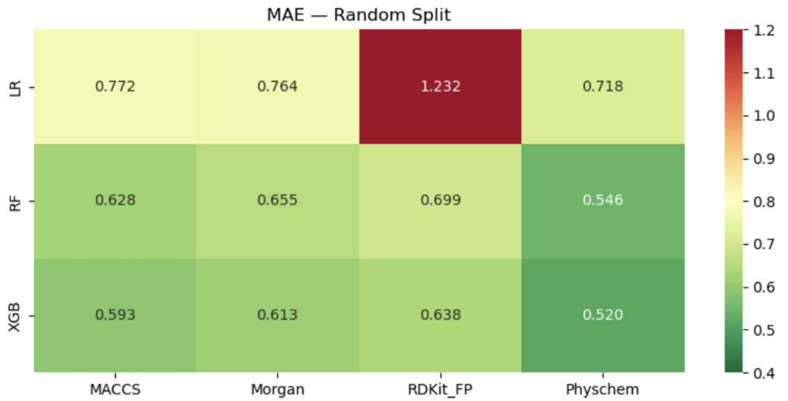
- Hyperparameters tuned on validation set only

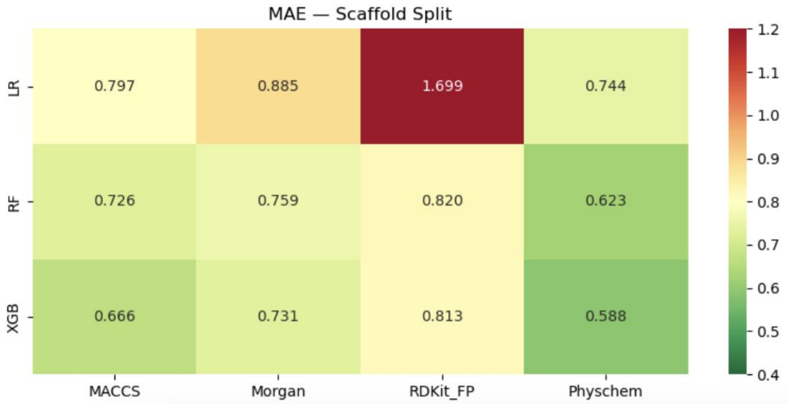
Model	Hyperparameters
Linear Regression	None — no hyperparameters
Random Forest	`n_estimators`, `max_depth`, `min_samples_leaf`
XGBoost	`n_estimators`, `max_depth`, `learning_rate`

Three metrics on test set: **MAE** (primary), **RMSE** (MoleculeNet comparison), **R^2** (variance explained)

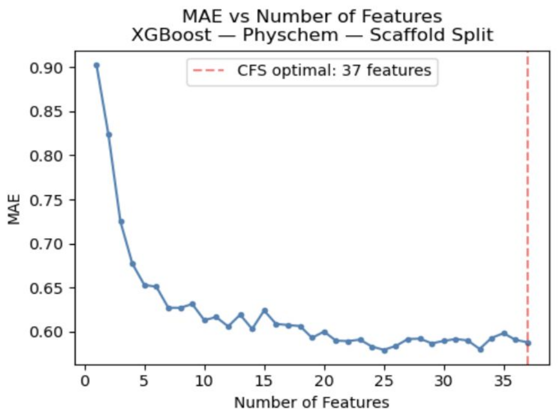
Consistent rankings across all combinations:

- **Models**: XGBoost > Random Forest > Linear Regression
- **Encodings**: Physchem > MACCS > Morgan > RDKit_FP
- **Split effect**: all models degrade on scaffold split — XGBoost most robust





Best Model, Feature & Learning Curve

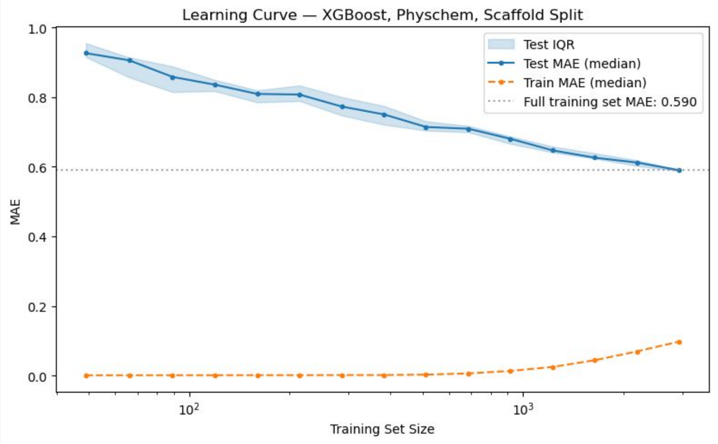



```
Best model per split (by MAE):
  Random: XGB_Physchem_Random — MAE=0.520, RMSE=0.732, R²=0.637
  Scaffold: XGB_Physchem_Scaffold — MAE=0.588, RMSE=0.765, R²=0.584
```

- Competitive with MoleculeNet GCN benchmark (RMSE~0.65, Wu et al. 2018) using only hand-crafted descriptor

Features:

- Performance plateaus after ~20–25 features
- Optimal by test MAE: 25 features (MAE=0.580) vs 37 CFS-selected (MAE=0.588) — negligible difference



Learning curve:

- MAE: 0.927 (n=49) → 0.590 (n=2,939)
- Curve not plateaued → **data-limited regime**

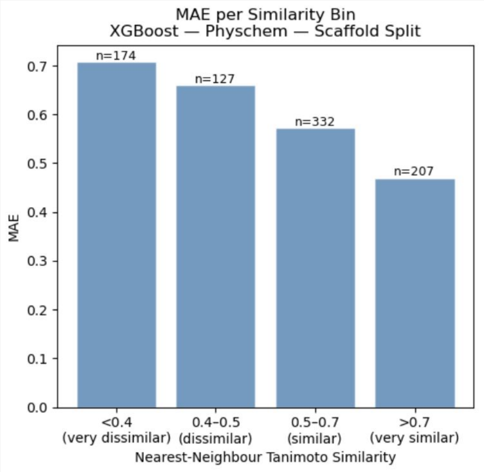
Similarity Analysis & Applicability Domain

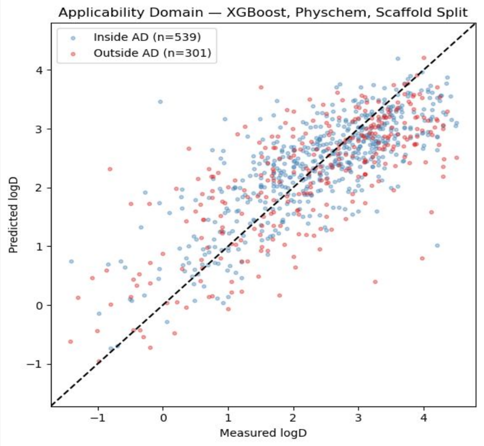
Performance degrades as molecules become more dissimilar from training set

```
Bin                              N      MAE      RMSE        R²
---------------------------------------------------------------
<0.4 (very dissimilar)         174    0.707     0.921     0.565
0.4-0.5 (dissimilar)           127    0.660     0.824     0.535
0.5-0.7 (similar)              332    0.572     0.750     0.583
>0.7 (very similar)            207    0.470     0.587     0.655
```

Applicability domain — threshold nn_sim = 0.5 (empirically determined — largest single MAE drop):

- Inside AD (64.2%): MAE = 0.533
- Outside AD (35.8%): MAE = 0.687 → **29% increase**



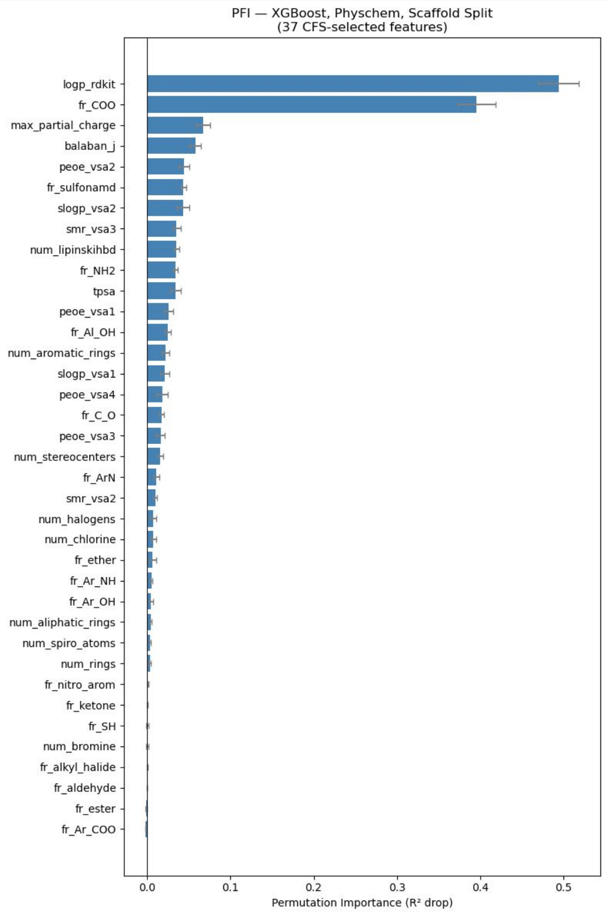


Explainability (PFI)

Top features:

1. `logp_rdkit` (0.494) — closest proxy to logD, logP and logD both measure lipophilicity
2. `fr_COO` (0.396) — carboxylic acid groups reduce lipophilicity by increasing polarity
3. `max_partial_charge` (0.067) — electronic properties influence ionization
4. Remaining: gradual decrease to near zero

CFS ensured no substitution effect → importance scores are chemically interpretable and reliable



Additional Graph Models

GCN (Graph Convolutional Network)

&

GIN (Graph Isomorphism Network)

We also decided to train 2 different type of graph models:

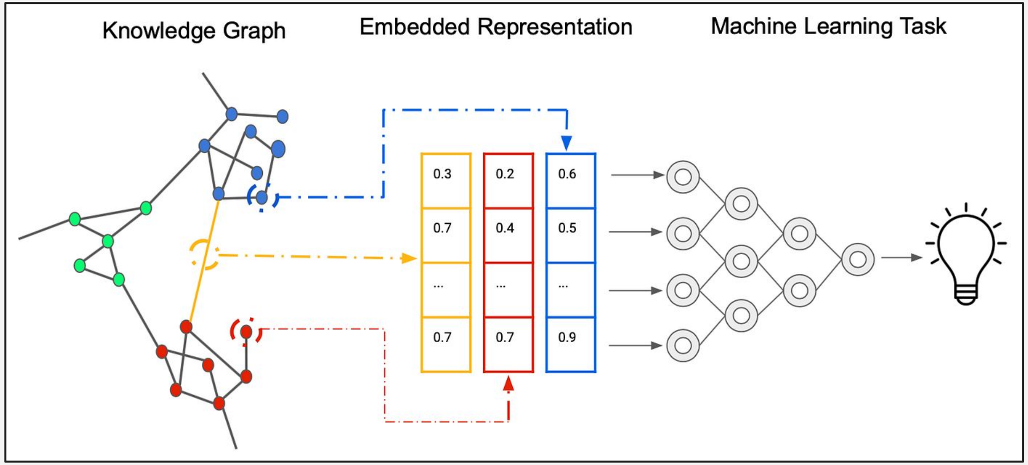
Popular in drug design environments.

Why? -> Molecules are graphs.

Atoms are nodes
Chemical bonds are edges

Capture geometry and learn patterns

Beyond traditional linear data

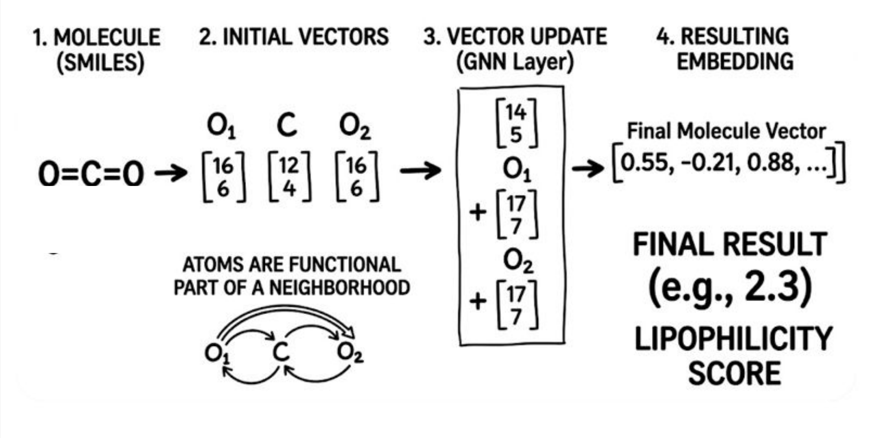


Additional Graph Models

Each atom (node) talks to its neighbors to figure out its own identity (embedding).

Atoms become a functional part of a neighborhood of atoms.

Bonds become numerical filters that help shape the identity of atoms.



< Very simplified explanation

GCN vs GIN

GCN	GIN
Updates atom vectors by aggregating the mean average of neighbors. (loses track of exact bond counts)	Updates atom vectors by aggregating the sum of neighbors (preserves the exact local structure)
Uses a single linear learnable weight matrix layer.	Uses Multi-Layer-Perceptron (MLP)

GIN is better for predicting lipophilicity because it preserves molecular size and atom counts

In addition, MLP helps in finding complex non-linear structural patterns

Graph Models

Scratch GCN:

A minimal, custom implementation using dense matrices and basic atomic features.

Batch size 1 (slow)
2 layers (128 hidden)
Early stopping

Best R2: 0.477

PyG GCN:

PyTorch Geometric implementation utilising rich OGB chemical features.

Batch size 64
2 GCNconv layers (128 hidden)
LR Decay
Early stopping

Best R2: 0.685

PyG GIN:

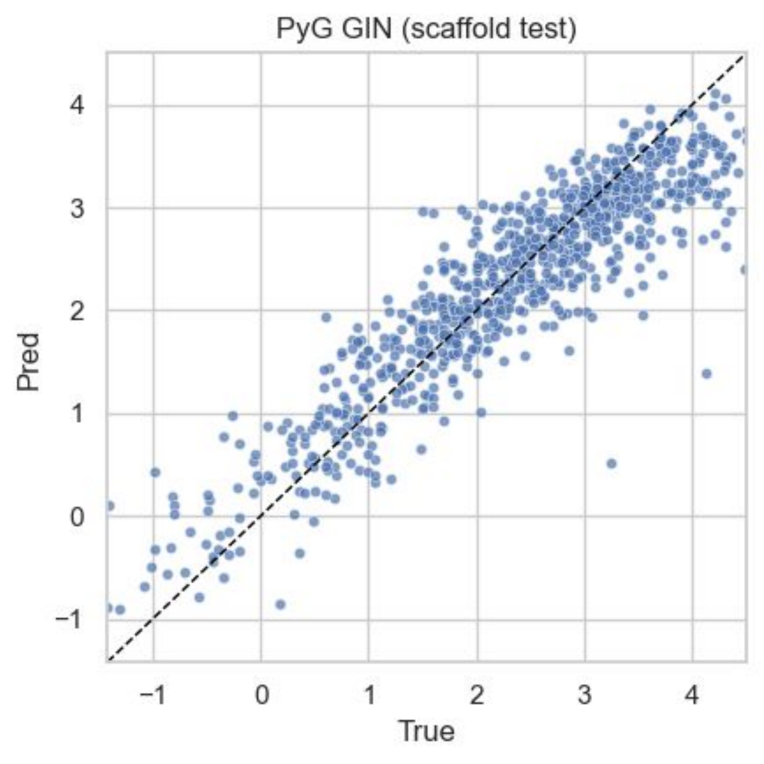
PyTorch Geometric impl. utilising rich OGB chemical features. Replacing simple averaging with MLP-based aggregation to better capture complex molecular structures

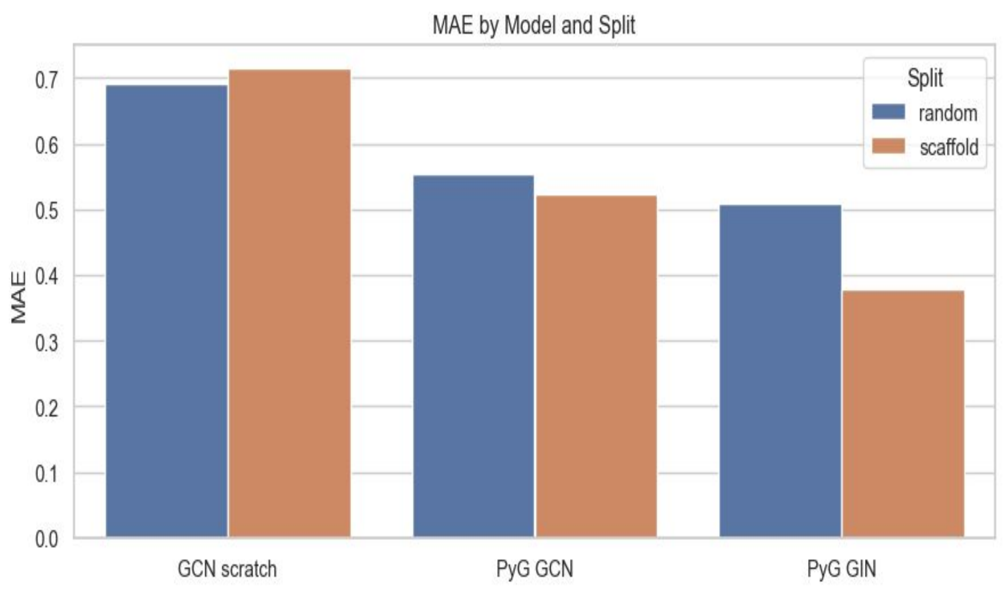
Batch size 64
2 GINconv layers (128 hidden)
LR Decay
Early stopping

Best R2: 0.822

Graph Models

	split	model	rmse	mae	r2
2	random	PyG GIN	0.667223	0.508073	0.698684
1	random	PyG GCN	0.712803	0.553830	0.656110
0	random	GCN scratch	0.878225	0.690881	0.477975
5	scaffold	PyG GIN	0.499992	0.378595	0.822537
4	scaffold	PyG GCN	0.665520	0.523527	0.685584
3	scaffold	GCN scratch	0.904276	0.713922	0.419524





Best Models

<u>Best non-graph model:</u>

XGBoost

MAE: 0.520
RMSE: 0.732
R2: 0.637

<u>Best graph model:</u>

GIN

MAE: 0.378
RMSE: 0.499
R2: 0.822

Comparison to existing models

Schroeter et al., 2007:
14’000 compounds

Gaussian Process regression model:
- RMSE: 0.66,
- MAE: 0.38

Support Vector regression model:
- RMSE: 0.71,
- MAE: 0.4

(R^2 score not mentioned)

Lenselink et al., 2021:
Multiple different datasets
(over 18’000 compounds, Opera datasets, ChEMBL_25, SAMPL7,...)

D-MPNN-based regression model:
- RMSE: 0.66,
- MAE: 0.48

Challenge D-MPNN model:
- RMSE: 0.35,
- R^2: 0.75

Conclusion

We were able to create several models capable of solving the problem given, lipophilicity (LogD7.4) prediction.

GNNs show particular promise, and are ever more widespread in use.

To obtain a more competitive model we can improve by following the experts:

- Improve evaluation strategies
- Use more data for training
- Explore more complex models (ensembles or hybrids, D-PMNNs,...) for robustness and generalisation
- Use multiple models to create additional features

https://beckybaileystudio.com/lipophilic-decorated-stone-skins/view/9244394/1/10524581 (image in intro)

https://github.com/ambiddisco/ML4DD-lipophilicity/tree/main (GitHub repo (check louie branch for graph models))

(used research papers in our repo)

Thank you for your attention!

Any [Q]u-[e]=[s]-[Ti]-[O]=n-[s] ?